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UNIVERSITÀ DEGLI STUDI DI PADOVA

From BBS to Blackboard: The Historical Pedagogy of Hacker Culture in Phrack Magazine (1985-2016)

Informal Learning and Identity in Hacker Subcultures

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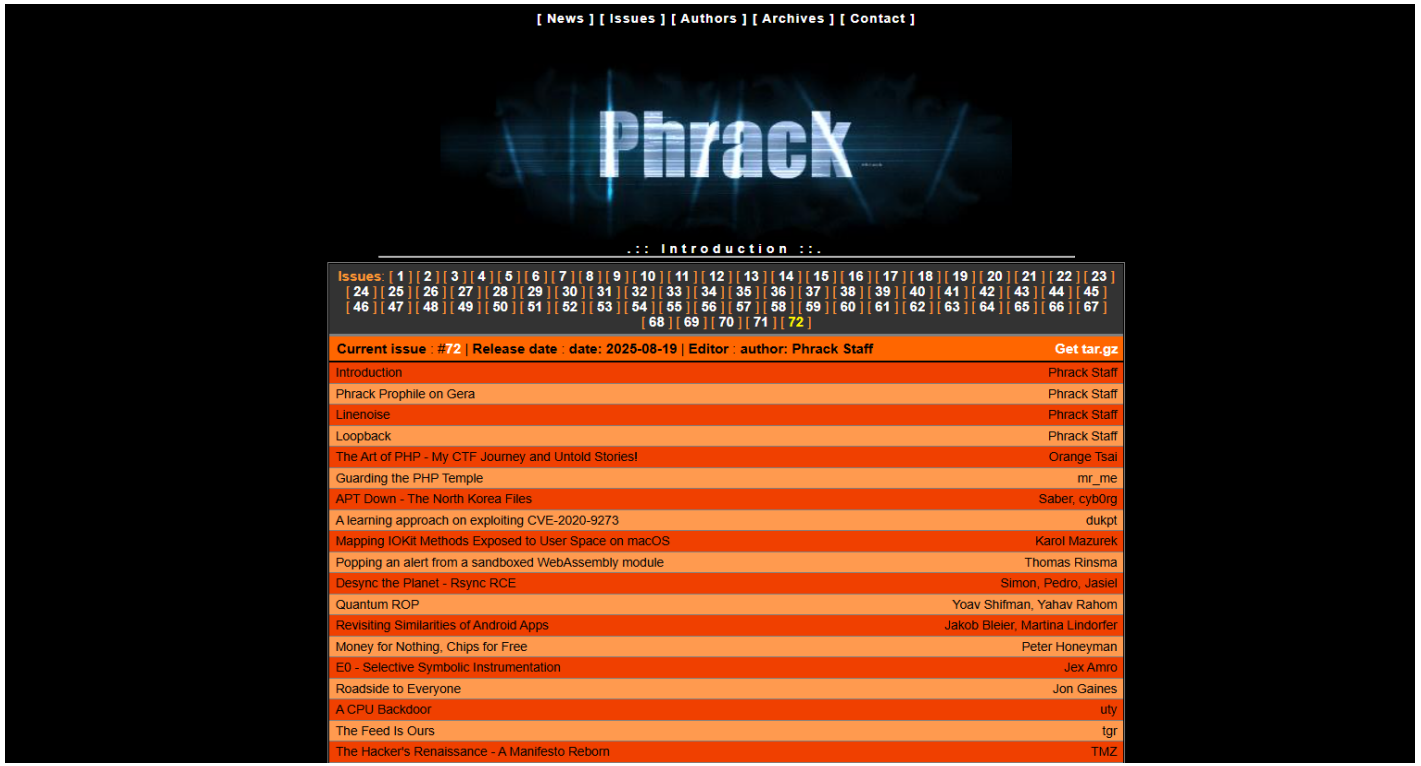


- Founded in 1985 on **Metal Shop BBS** by *Taran King* and *Knight Lightning*
- One of the most influential underground hacker e-zines
- Ran for over 30 years **(1985–2025)**, publishing 71 issues
- Combined technical tutorials, manifestos, editorials, and scene reports
- Written by hackers, for hackers
- Key site for sharing skills, ethics, and subcultural identity
- Served as a peer-based learning space and ideological platform
- **Now studied as a historical corpus of informal pedagogy**

Research Problem

How did *Phrack Magazine* (1985–2016) function as an informal pedagogical space where hacker culture, values, and identities were taught, negotiated, and reproduced?

Scholarly & Historical Gap

- Informal learning in hacker communities remains under-researched in historical and educational literature.
- Existing studies focus on policy, security, or media representation.
- This thesis shifts the focus to pedagogy

Background & Significance

- Hacker culture has long been viewed through legal, technical, or moral lenses.
- Dominant narratives focus on crime, security, or technological genius, but rarely consider how hackers learned, taught, and transmitted values.
- *Phrack*, a key underground publication, served not only as a communication platform but as a site of peer-to-peer learning, subcultural reproduction, and ideological training.

Research Questions

- How are peer-based learning practices constructed in *Phrack*?
- How are readers positioned as learners, mentors, or community members?
- How are hacker values (curiosity, resistance, sharing) expressed?
- Where are the boundaries of inclusion/exclusion?

Origins of Hacker Culture (1940s–1970s)

- **Post-WWII tech boom** → U.S. becomes tech hub.
- **Phone phreaking** in 1960s: hacking telephone systems.
- **MIT labs, Homebrew Computer Club (1975)**: birth of hacker ethics.

Shared values: curiosity, anti-authority, technical mastery.

From BBS to Phrack: Building the Hacker Classroom

- **Early 1980s:**

BBS (Bulletin Board Systems) emerge as decentralized, dial-up forums for file sharing, messaging, and software exchange.

- **1981–1984:**

BBSes become central to the "computer underground", offering hackers spaces to share knowledge anonymously and build pseudonymous reputations.

- **November 17, 1985:**

Phrack Magazine launches on **Metal Shop BBS**, founded by Taran King and Knight Lightning.

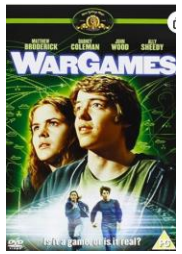
- **1985–onward:**

Phrack functions as a hacker classroom:

- Technical tutorials
- Ethical manifestos
- Community debates
- Identity formation

From Curiosity to Criminality: Media Panic & State Crackdowns

- **1983 – *WarGames* Released:**



- **Mid–Late 1980s – Media Narratives Shift:**
  Hackers increasingly framed as cybercriminals, deviant youth, and threats to national security.
- **1986 – *The Conscience of a Hacker* Published:**
  Written by *The Mentor* after his arrest.
  → Becomes the **manifesto** of misunderstood, intelligent youth.
  → *"The true crime of the hacker is curiosity."*
- **1990 – Operation Sundevil:**
  Large-scale law enforcement raid targeting U.S. hackers.
  → Phrack responds with legal self-help guides and more politicized content.
  → Shift from celebration to survival rhetoric.

From Theory to Practice: Informal Learning in the Digital Age

Informal Learning is non-institutional, socially embedded, and learner-driven (Vygotsky, 1934; Rogoff, 1990). It is characterized by:

1. Non-didactic
2. Socially interactive
3. Embedded in activity that is meaningful
4. Triggered by the learner's interests or needs
5. Typically free of formal evaluation

Core Theoretical Foundations

- **John Dewey** *(1916)*

Education as experiential, democratic, and integrated into everyday life.
↳ Hacker pedagogy reflects Dewey's model of learning through doing.

- **Lev Vygotsky** *(1934)*

Importance of everyday concepts as foundations for scientific knowledge.
↳ Informal tech experimentation in hacker culture as pre-formal cognitive base.

- **Jean Lave & Etienne Wenger (1991) – Communities of Practice**

Lave & Wenger argue that learning is fundamentally a social process. It doesn't happen only through formal instruction, but by participating in communities of practice.

- **Richard Stallman(1980s–2001)– Hacker Ethic**

Advocated for openness, anti-authority, and information freedom in computing.

- **Steven Levy (1984) – Hackers: Heroes of the Computer Revolution**

Documented core hacker values: curiosity, mastery, resistance to control.

- **Dick Hebdige (1979) – Subculture and Style**

Subcultures as sites of symbolic resistance through style and appropriation.

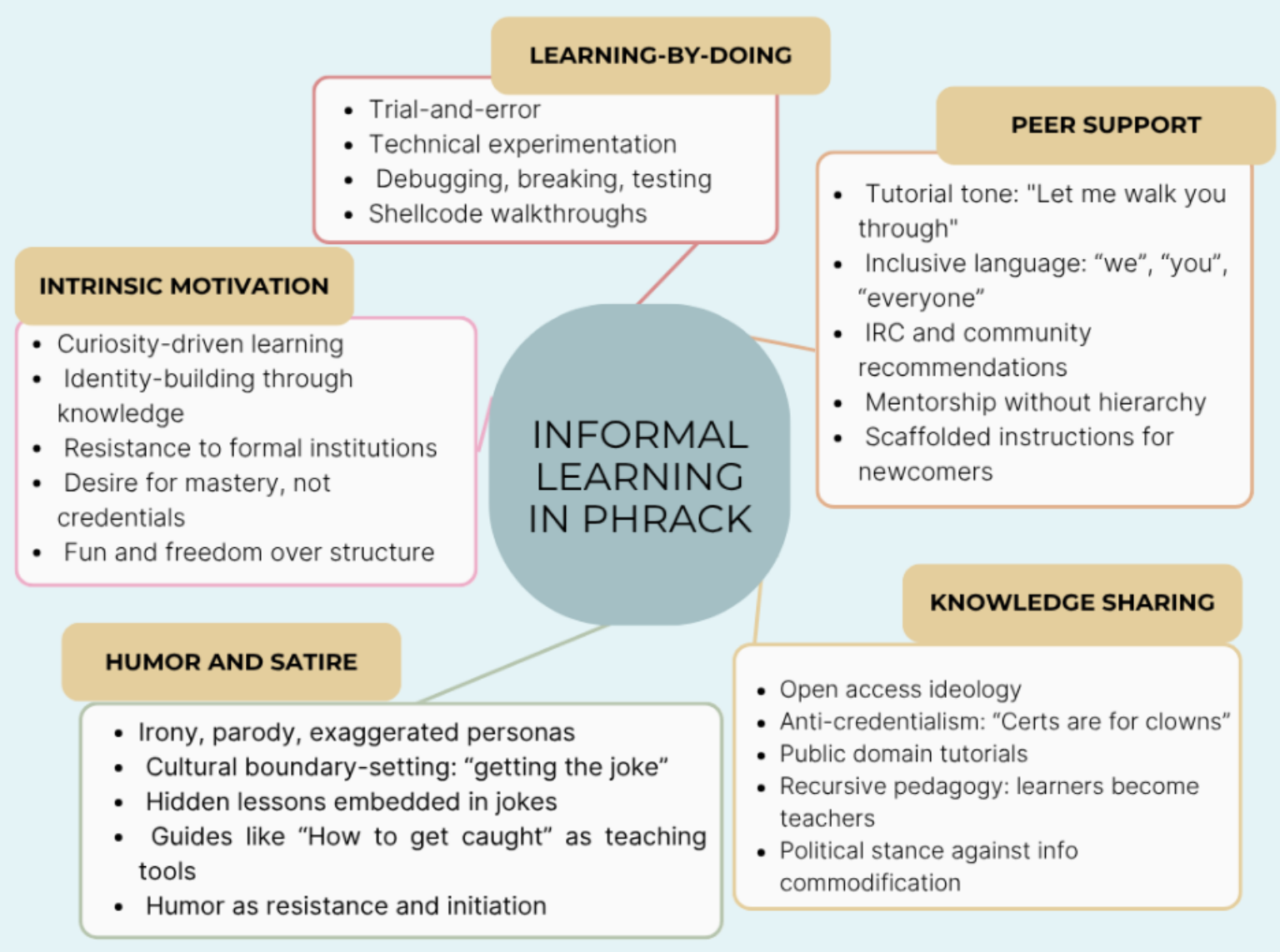
- **Sarah Thornton (1995) – Subcultural Capital**

Introduced idea of subcultural status through insider knowledge and practices.

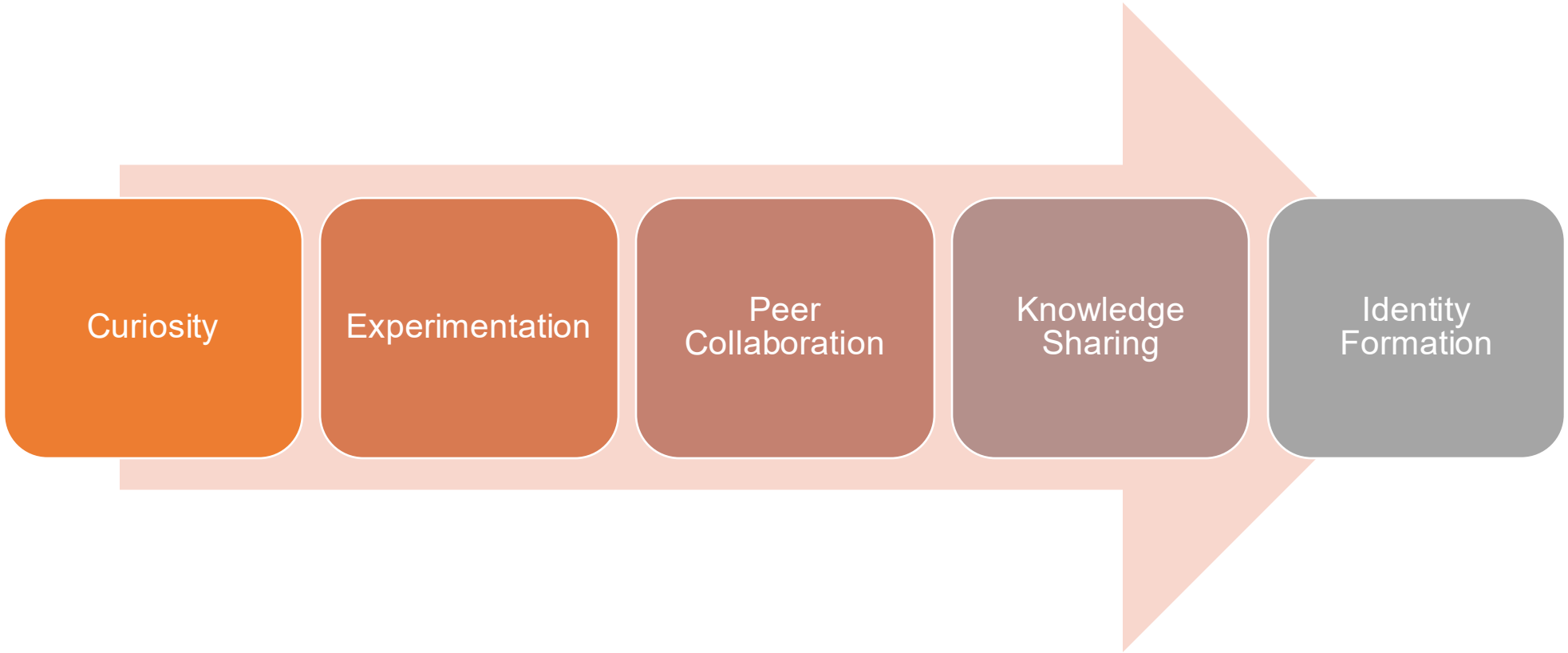
Methodology

- Corpus: 69 issues (1985–2016)
- Tools: Voyant + CDA
- Genres: tutorials, manifestos, editorials, scene reports
- Mixed method = quantitative patterns + qualitative depth

Key Findings: Modes of Learning in *Phrack*



Modes of Learning in Phrack



Gendered Absence in a Historical Male-Dominated Space

8	woman*	82

Distribution:

Collocates:
said (5); pwn (4); named (4); behind (4); year (3)

Correlations:
computer (6665); people (4389); pwn (4246); data (6536); like (5504)

Phrases:
"woman behind the front desk"
"womanizer one of comsex's founders"
"woman behind the front"

Items:

	Term	Count	Trend
1	men*	1943	

Distribution:

Collocates:
menu (99); people (69); like (44); information (38); options (36)

Correlations:
computer (6665); information (6763); like (5504); data (6536); wed (12364)

Phrases:
"mentioning or referencing this software or this function license is also granted to make and use derivative works provided that such works are identified as derived from the rsa data security inc md5 message digest algorithm in all material mentioning or referencing the derived work rsa data security inc makes no representations concerning either the merchantability of this software or the suitability of this software for any particular purpose it is provided as is without express or implied warranty of any kind these notices must be retained in any copies of any part of this documentation and or software"

Items:

Limitations & Ethics

- **Focus on Anglophone, male-dominated discourse**
- **Lack of ethnographic access** → no interviews with editors/contributors
- **Potential bias in CDA** → interpretive framework, not objective measurement
- **Historical distance** → treated archives as cultural artifacts, not legal evidence
- **Phrack as a partial archive** → cannot represent the full hacker community

Conclusion

Aspect	Phrack Magazine	Formal Education
Learning model	Peer-to-peer, informal	Instructor-led, structured
Motivation	Curiosity, autonomy	Certification, career
Access	Free, open, remixable	Restricted, institutional
Authority	Distributed, self-built	Centralized, credentialed
Assessment	*Reputation, contribution*	*Exams, grades*

Bibliography

https://drive.google.com/file/d/1iFEOqNwNCVk5ywF6KLy17HMjTbWVn7J_/view?usp=drive_link

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